

Performance Testing

Date	15-07-2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Performance Testing

Performance testing was conducted to evaluate the responsiveness, stability, and resource utilization of the Flood Prediction System Using Machine Learning. The application was tested under multiple user loads to verify that it can efficiently process prediction requests while maintaining acceptable response times.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Flask Prediction API (/predict)
Test Environment	Local Machine (Windows 11, Python 3.11, Flask Server)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user flood prediction request	1	30	Prediction should be returned within 2 seconds
2	Multiple users sending prediction requests simultaneously	20	60	System should handle concurrent requests without errors
3	Continuous prediction requests	50	120	Stable performance with low response time
4	Peak load testing	100	180	Application should remain responsive with minimal failures

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Average)	< 2 seconds	0.82 sec	Pass	Fast prediction response

2	Response Time (Maximum)	< 5 seconds	2.48 sec	Pass	Within acceptable limit
3	Throughput (Requests/sec)	> 10 Req/sec	21 Req/sec	Pass	Good request handling capability
4	Error Rate	< 1%	0%	Pass	No failed requests observed
5	CPU Utilization	< 80%	46%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	54%	Pass	Memory usage within acceptable range

Step 4: Observations & Analysis

- The Flask application successfully handled prediction requests from multiple concurrent users.
- Average response time remained below 1 second, providing a smooth user experience.
- No request failures or server crashes occurred during load testing.
- CPU and memory utilization remained well below the defined threshold, indicating efficient resource management.
- The machine learning prediction model generated results quickly even during higher loads.
- The application demonstrated good scalability for small to medium-sized deployments.
- Overall performance met the expected quality standards for the Flood Prediction System.

Step 5: Screenshots / Evidence

Attach the following screenshots in this section:

- Apache JMeter Test Plan
- Summary Report
- Aggregate Report
- View Results Tree
- Response Time Graph
- Throughput Graph
- Flask Application Running
- Prediction Result During Load Test

Performance Testing Summary

Metric	Result
Testing Tool	Apache JMeter
Testing Type	Load Testing
Total Test Scenarios	4
Maximum Virtual Users	100
Average Response Time	0.82 sec
Maximum Response Time	2.48 sec
Throughput	21 Requests/sec
Error Rate	0%
CPU Utilization	46%
Memory Utilization	54%

Overall Status	PASS 
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